Bayesian Inference and Kalman Filtering for the Stochastic Damped Harmonic Oscillator

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Abstract

We develop a complete Bayesian framework for state estimation of the stochastic damped harmonic oscillator governed by the Langevin equation $x''(t) + ax'(t) + bx(t) = \sigma\eta(t)$. Starting from the exact time-dependent solution of the Fokker-Planck equation, we derive the prior distribution for the system state. Combining this with a Gaussian measurement model via Bayes' theorem yields a posterior distribution that remains Gaussian, leading to explicit Kalman filter update equations. We provide the complete mathematical derivation of the prediction and update steps, including explicit formulas for the Kalman gain, posterior mean, and posterior covariance matrix for position-only, velocity-only, and full-state measurements. We also derive the Bayesian evidence (marginal likelihood) and its use in parameter estimation and model selection. The framework presented here provides a rigorous mathematical foundation for data assimilation in stochastic dynamical systems.

Keywords: Bayesian inference, Kalman filter, stochastic differential equations, harmonic oscillator, data assimilation, Gaussian processes
MSC Classifications: 62F15, 60J60, 93E11, 60H10

1 Introduction

The stochastic damped harmonic oscillator

$$x''(t) + ax'(t) + bx(t) = \sigma\eta(t), \tag{1}$$

where $\eta(t)$ is Gaussian white noise, provides a fundamental model for numerous physical systems. In many practical applications, we do not have direct access to the full state $\mathbf{X}(t) = (x(t), v(t))^T$. Instead, we make noisy measurements at discrete times:

$$\mathbf{Y}_t = H\mathbf{X}(t) + \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(0, R). \tag{2}$$

The problem of estimating the true state from noisy measurements is fundamental to stochastic control, signal processing, and experimental physics [1, 2, 3]. Bayesian inference provides a principled framework for combining our dynamical model (the prior) with observational data (the likelihood) to obtain an updated belief (the posterior).

In this paper, we develop the complete Bayesian framework for state estimation of the stochastic damped harmonic oscillator. The linearity of the dynamics and the Gaussianity of both the process noise and measurement noise lead to a conjugate Gaussian update, resulting in the classical Kalman filter [4]. However, our emphasis is on the mathematical derivation from first principles, making explicit use of the time-dependent covariance matrix derived in our companion paper.

2 Mathematical Preliminaries

2.1 State-Space Representation

We rewrite (1) in state-space form:

$$d\mathbf{X}(t) = A\mathbf{X}(t)\,dt + B\,dW(t), \tag{3}$$

where

$$\mathbf{X}(t) = \begin{pmatrix} x(t) \\ v(t) \end{pmatrix}, \quad A = \begin{pmatrix} 0 & 1 \\ -b & -a \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ \sigma \end{pmatrix}. \tag{4}$$

The discrete-time transition over an interval Δt is:

$$\mathbf{X}_t = \Phi\mathbf{X}_{t-1} + \boldsymbol{\xi}_t, \tag{5}$$

where

$$\Phi = e^{A\Delta t} \tag{6}$$

is the state transition matrix, and

$$\boldsymbol{\xi}_t \sim \mathcal{N}(0, Q(\Delta t)) \tag{7}$$

is the process noise with covariance:

$$Q(\Delta t) = \int_0^{\Delta t} e^{A\tau} BB^T e^{A^T\tau}\,d\tau. \tag{8}$$

The matrix $Q(\Delta t)$ is precisely the time-dependent covariance matrix derived in our companion paper:

$$Q(\Delta t) = \begin{pmatrix} Q_{11} & Q_{12} \\ Q_{12} & Q_{22} \end{pmatrix}. \tag{9}$$

2.2 The Transition Density

From (5), the transition density is Gaussian:

$$\boxed{p(\mathbf{X}_t|\mathbf{X}_{t-1}) = \mathcal{N}(\mathbf{X}_t; \Phi\mathbf{X}_{t-1}, Q(\Delta t)).} \tag{10}$$

2.3 The Measurement Model

At each time step, we observe:

$$\mathbf{Y}_t = H\mathbf{X}_t + \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(0, R). \tag{11}$$

The likelihood function is:

$$\boxed{p(\mathbf{Y}_t|\mathbf{X}_t) = \mathcal{N}(\mathbf{Y}_t; H\mathbf{X}_t, R).} \tag{12}$$

3 The Bayesian Filtering Framework

3.1 Bayes' Theorem for Sequential State Estimation

The fundamental equation of Bayesian filtering is:

$$\boxed{p(\mathbf{X}_t|\mathbf{Y}_{1:t}) = \frac{p(\mathbf{Y}_t|\mathbf{X}_t)\, p(\mathbf{X}_t|\mathbf{Y}_{1:t-1})}{p(\mathbf{Y}_t|\mathbf{Y}_{1:t-1})}.} \tag{13}$$

The denominator is the marginal likelihood (evidence):

$$p(\mathbf{Y}_t|\mathbf{Y}_{1:t-1}) = \int p(\mathbf{Y}_t|\mathbf{X}_t)\, p(\mathbf{X}_t|\mathbf{Y}_{1:t-1})\, d\mathbf{X}_t. \tag{14}$$

3.2 The Prior Distribution

The prior is obtained by propagating the previous posterior through the transition density:

$$p(\mathbf{X}_t|\mathbf{Y}_{1:t-1}) = \int p(\mathbf{X}_t|\mathbf{X}_{t-1})\, p(\mathbf{X}_{t-1}|\mathbf{Y}_{1:t-1})\, d\mathbf{X}_{t-1}. \tag{15}$$

Since both the transition density and the previous posterior are Gaussian, the prior is Gaussian:

$$\boxed{p(\mathbf{X}_t|\mathbf{Y}_{1:t-1}) = \mathcal{N}(\mathbf{X}_t; \boldsymbol{\mu}_t^-, \Sigma_t^-).} \tag{16}$$

4 The Prediction Step (Prior)

4.1 Mean Prediction

The predicted mean is obtained by evolving the previous estimate through the deterministic dynamics:

$$\boxed{\boldsymbol{\mu}_t^- = \Phi\boldsymbol{\mu}_{t-1}^+ = e^{A\Delta t}\boldsymbol{\mu}_{t-1}^+.} \tag{17}$$

Proof.

$$\boldsymbol{\mu}_t^- = \mathbb{E}[\mathbf{X}_t|\mathbf{Y}_{1:t-1}] = \mathbb{E}[\Phi\mathbf{X}_{t-1} + \boldsymbol{\xi}_t|\mathbf{Y}_{1:t-1}] = \Phi\mathbb{E}[\mathbf{X}_{t-1}|\mathbf{Y}_{1:t-1}] = \Phi\boldsymbol{\mu}_{t-1}^+.$$

□

4.2 Covariance Prediction

The predicted covariance is:

$$\boxed{\Sigma_t^- = \Phi\Sigma_{t-1}^+\Phi^T + Q(\Delta t).} \tag{18}$$

Proof.

$$\Sigma_t^- = \mathrm{Cov}[\mathbf{X}_t|\mathbf{Y}_{1:t-1}] = \mathrm{Cov}[\Phi\mathbf{X}_{t-1}+\boldsymbol{\xi}_t|\mathbf{Y}_{1:t-1}] = \Phi\,\mathrm{Cov}[\mathbf{X}_{t-1}|\mathbf{Y}_{1:t-1}]\Phi^T+\mathrm{Cov}[\boldsymbol{\xi}_t] = \Phi\Sigma_{t-1}^+\Phi^T+Q(\Delta t).$$

□

4.3 The Innovation Covariance

The covariance of the innovation $\mathbf{r}_t = \mathbf{Y}_t - H\boldsymbol{\mu}_t^-$ is:

$$\boxed{S_t = H\Sigma_t^- H^T + R.} \tag{19}$$

This matrix will be essential for computing the Kalman gain.

5 The Update Step (Posterior)

5.1 Product of Gaussians

The prior and likelihood are both Gaussian:

$$p(\mathbf{X}_t|\mathbf{Y}_{1:t-1}) \propto \exp\left[-\frac{1}{2}(\mathbf{X}_t - \boldsymbol{\mu}_t^-)^T(\Sigma_t^-)^{-1}(\mathbf{X}_t - \boldsymbol{\mu}_t^-)\right],$$

$$p(\mathbf{Y}_t|\mathbf{X}_t) \propto \exp\left[-\frac{1}{2}(\mathbf{Y}_t - H\mathbf{X}_t)^T R^{-1}(\mathbf{Y}_t - H\mathbf{X}_t)\right].$$

Their product is proportional to:

$$\exp\left[-\frac{1}{2}\left((\mathbf{X}_t - \boldsymbol{\mu}_t^-)^T(\Sigma_t^-)^{-1}(\mathbf{X}_t - \boldsymbol{\mu}_t^-) + (\mathbf{Y}_t - H\mathbf{X}_t)^T R^{-1}(\mathbf{Y}_t - H\mathbf{X}_t)\right)\right].$$

5.2 Completing the Square

We write the exponent in the form:

$$-\frac{1}{2}\left[\mathbf{X}_t^T(\Sigma_t^+)^{-1}\mathbf{X}_t - 2\mathbf{X}_t^T(\Sigma_t^+)^{-1}\boldsymbol{\mu}_t^+ + \text{constant}\right],$$

where the posterior covariance and mean are:

$$\boxed{(\Sigma_t^+)^{-1} = (\Sigma_t^-)^{-1} + H^T R^{-1} H,} \tag{20}$$

$$\boxed{(\Sigma_t^+)^{-1}\boldsymbol{\mu}_t^+ = (\Sigma_t^-)^{-1}\boldsymbol{\mu}_t^- + H^T R^{-1}\mathbf{Y}_t.} \tag{21}$$

Therefore:

$$\boxed{\Sigma_t^+ = \left[(\Sigma_t^-)^{-1} + H^T R^{-1} H\right]^{-1},} \tag{22}$$

$$\boxed{\boldsymbol{\mu}_t^+ = \Sigma_t^+\left[(\Sigma_t^-)^{-1}\boldsymbol{\mu}_t^- + H^T R^{-1}\mathbf{Y}_t\right].} \tag{23}$$

5.3 The Kalman Gain

The Kalman gain matrix relates the posterior mean to the innovation:

$$\boldsymbol{\mu}_t^+ = \boldsymbol{\mu}_t^- + K_t(\mathbf{Y}_t - H\boldsymbol{\mu}_t^-). \tag{24}$$

Substituting (22)-(23) into (24) and solving for K_t:

$$\boxed{K_t = \Sigma_t^- H^T(H\Sigma_t^- H^T + R)^{-1}.} \tag{25}$$

Proof. Starting from (23):

$$\boldsymbol{\mu}_t^+ = \Sigma_t^+ \left[(\Sigma_t^-)^{-1}\boldsymbol{\mu}_t^- + H^T R^{-1}\mathbf{Y}_t \right].$$

Add and subtract $H^T R^{-1} H\boldsymbol{\mu}_t^-$:

$$= \Sigma_t^+ \left[(\Sigma_t^-)^{-1}\boldsymbol{\mu}_t^- + H^T R^{-1} H\boldsymbol{\mu}_t^- - H^T R^{-1} H\boldsymbol{\mu}_t^- + H^T R^{-1}\mathbf{Y}_t \right].$$

$$= \Sigma_t^+ \left[(\Sigma_t^+)^{-1}\boldsymbol{\mu}_t^- + H^T R^{-1}(\mathbf{Y}_t - H\boldsymbol{\mu}_t^-) \right].$$

$$= \boldsymbol{\mu}_t^- + \Sigma_t^+ H^T R^{-1}(\mathbf{Y}_t - H\boldsymbol{\mu}_t^-).$$

Therefore:

$$K_t = \Sigma_t^+ H^T R^{-1}.$$

Using the matrix inversion lemma (Appendix A):

$$\Sigma_t^+ H^T R^{-1} = \Sigma_t^- H^T (H\Sigma_t^- H^T + R)^{-1}.$$

Thus:

$$K_t = \Sigma_t^- H^T (H\Sigma_t^- H^T + R)^{-1}.$$

□

5.4 Posterior Covariance in Innovation Form

The posterior covariance can also be written as:

$$\boxed{\Sigma_t^+ = (I - K_t H)\Sigma_t^-.} \tag{26}$$

Proof.

$$\Sigma_t^+ = \left[(\Sigma_t^-)^{-1} + H^T R^{-1} H \right]^{-1}.$$

Using the matrix inversion lemma (Appendix A):

$$\Sigma_t^+ = \Sigma_t^- - \Sigma_t^- H^T (H\Sigma_t^- H^T + R)^{-1} H\Sigma_t^-.$$

$$= \Sigma_t^- - K_t H\Sigma_t^- = (I - K_t H)\Sigma_t^-.$$

□

6 Explicit Formulas for Common Observation Types

6.1 Position-Only Measurements

We observe only position:

$$H = \begin{pmatrix} 1 & 0 \end{pmatrix}, \quad R = \sigma_y^2.$$

6.1.1 Innovation Covariance

$$S_t = H\Sigma_t^- H^T + R = \Sigma_{11,t}^- + \sigma_y^2. \tag{27}$$

6.1.2 Kalman Gain

$$K_t = \Sigma_t^- H^T (H\Sigma_t^- H^T + R)^{-1} = \frac{1}{\Sigma_{11,t}^- + \sigma_y^2} \begin{pmatrix} \Sigma_{11,t}^- \\ \Sigma_{12,t}^- \end{pmatrix}. \tag{28}$$

6.1.3 Mean Update

$$\boxed{\mu_{x,t}^+ = \mu_{x,t}^- + \frac{\Sigma_{11,t}^-}{\Sigma_{11,t}^- + \sigma_y^2}(Y_t - \mu_{x,t}^-),} \tag{29}$$

$$\boxed{\mu_{v,t}^+ = \mu_{v,t}^- + \frac{\Sigma_{12,t}^-}{\Sigma_{11,t}^- + \sigma_y^2}(Y_t - \mu_{x,t}^-).} \tag{30}$$

6.1.4 Covariance Update

$$\boxed{\Sigma_{11,t}^+ = \frac{\Sigma_{11,t}^- \sigma_y^2}{\Sigma_{11,t}^- + \sigma_y^2},} \tag{31}$$

$$\boxed{\Sigma_{12,t}^+ = \frac{\Sigma_{12,t}^- \sigma_y^2}{\Sigma_{11,t}^- + \sigma_y^2},} \tag{32}$$

$$\boxed{\Sigma_{22,t}^+ = \Sigma_{22,t}^- - \frac{(\Sigma_{12,t}^-)^2}{\Sigma_{11,t}^- + \sigma_y^2}.} \tag{33}$$

Remark 6.1. *The position variance is always reduced by the measurement. The velocity variance is also reduced through its correlation with position.*

6.2 Velocity-Only Measurements

We observe only velocity:

$$H = \begin{pmatrix} 0 & 1 \end{pmatrix}, \quad R = \sigma_y^2.$$

6.2.1 Innovation Covariance

$$S_t = \Sigma_{22,t}^- + \sigma_y^2. \tag{34}$$

6.2.2 Kalman Gain

$$K_t = \frac{1}{\Sigma_{22,t}^- + \sigma_y^2} \begin{pmatrix} \Sigma_{12,t}^- \\ \Sigma_{22,t}^- \end{pmatrix}. \tag{35}$$

6.2.3 Mean Update

$$\boxed{\mu_{x,t}^+ = \mu_{x,t}^- + \frac{\Sigma_{12,t}^-}{\Sigma_{22,t}^- + \sigma_y^2}(Y_t - \mu_{v,t}^-),} \tag{36}$$

$$\boxed{\mu_{v,t}^+ = \mu_{v,t}^- + \frac{\Sigma_{22,t}^-}{\Sigma_{22,t}^- + \sigma_y^2}(Y_t - \mu_{v,t}^-).} \tag{37}$$

6.2.4 Covariance Update

$$\boxed{\Sigma_{11,t}^{+} = \Sigma_{11,t}^{-} - \frac{(\Sigma_{12,t}^{-})^2}{\Sigma_{22,t}^{-} + \sigma_y^2},} \tag{38}$$

$$\boxed{\Sigma_{12,t}^{+} = \frac{\Sigma_{12,t}^{-}\sigma_y^2}{\Sigma_{22,t}^{-} + \sigma_y^2},} \tag{39}$$

$$\boxed{\Sigma_{22,t}^{+} = \frac{\Sigma_{22,t}^{-}\sigma_y^2}{\Sigma_{22,t}^{-} + \sigma_y^2}.} \tag{40}$$

6.3 Full State Measurements

We observe both position and velocity with independent noise:

$$H = I_2, \quad R = \begin{pmatrix} \sigma_x^2 & 0 \\ 0 & \sigma_v^2 \end{pmatrix}.$$

6.3.1 Kalman Gain

$$K_t = \Sigma_t^{-}(\Sigma_t^{-} + R)^{-1}. \tag{41}$$

Since $H = I$, the update decouples somewhat.

6.3.2 Mean Update

$$\boxed{\mu_{x,t}^{+} = \frac{\mu_{x,t}^{-}\sigma_x^2 + Y_x\Sigma_{11,t}^{-}}{\Sigma_{11,t}^{-} + \sigma_x^2},} \tag{42}$$

$$\boxed{\mu_{v,t}^{+} = \frac{\mu_{v,t}^{-}\sigma_v^2 + Y_v\Sigma_{22,t}^{-}}{\Sigma_{22,t}^{-} + \sigma_v^2}.} \tag{43}$$

6.3.3 Covariance Update

$$\boxed{\Sigma_{11,t}^{+} = \frac{\Sigma_{11,t}^{-}\sigma_x^2}{\Sigma_{11,t}^{-} + \sigma_x^2},} \tag{44}$$

$$\boxed{\Sigma_{22,t}^{+} = \frac{\Sigma_{22,t}^{-}\sigma_v^2}{\Sigma_{22,t}^{-} + \sigma_v^2},} \tag{45}$$

$$\boxed{\Sigma_{12,t}^{+} = \frac{\Sigma_{12,t}^{-}\sigma_x^2\sigma_v^2}{(\Sigma_{11,t}^{-} + \sigma_x^2)(\Sigma_{22,t}^{-} + \sigma_v^2)}.} \tag{46}$$

Remark 6.2. *The cross-covariance is reduced by both measurements. In the limit of perfect measurements* $(\sigma_x^2 \to 0, \sigma_v^2 \to 0)$*, the posterior covariance vanishes.*

7 Bayesian Evidence (Marginal Likelihood)

7.1 Definition

The Bayesian evidence (marginal likelihood) is:

$$\boxed{Z_t = p(\mathbf{Y}_t|\mathbf{Y}_{1:t-1}) = \int p(\mathbf{Y}_t|\mathbf{X}_t)\, p(\mathbf{X}_t|\mathbf{Y}_{1:t-1})\, d\mathbf{X}_t.} \tag{47}$$

7.2 Explicit Form

Since both the prior and likelihood are Gaussian, the evidence is Gaussian:

$$\boxed{Z_t = \mathcal{N}(\mathbf{Y}_t; H\boldsymbol{\mu}_t^-, H\Sigma_t^- H^T + R).} \tag{48}$$

7.3 Position-Only Measurements

$$\boxed{Z_t = \frac{1}{\sqrt{2\pi(\Sigma_{11,t}^- + \sigma_y^2)}} \exp\left[-\frac{(Y_t - \mu_{x,t}^-)^2}{2(\Sigma_{11,t}^- + \sigma_y^2)}\right].} \tag{49}$$

7.4 Velocity-Only Measurements

$$\boxed{Z_t = \frac{1}{\sqrt{2\pi(\Sigma_{22,t}^- + \sigma_y^2)}} \exp\left[-\frac{(Y_t - \mu_{v,t}^-)^2}{2(\Sigma_{22,t}^- + \sigma_y^2)}\right].} \tag{50}$$

7.5 Full State Measurements

$$\boxed{Z_t = \frac{1}{2\pi\sqrt{(\Sigma_{11,t}^- + \sigma_x^2)(\Sigma_{22,t}^- + \sigma_v^2) - (\Sigma_{12,t}^-)^2}} \exp\left[-\frac{1}{2}\mathbf{r}_t^T S_t^{-1}\mathbf{r}_t\right],} \tag{51}$$

where

$$\mathbf{r}_t = \begin{pmatrix} Y_x - \mu_{x,t}^- \\ Y_v - \mu_{v,t}^- \end{pmatrix}, \quad S_t = \begin{pmatrix} \Sigma_{11,t}^- + \sigma_x^2 & \Sigma_{12,t}^- \\ \Sigma_{12,t}^- & \Sigma_{22,t}^- + \sigma_v^2 \end{pmatrix}.$$

8 Parameter Estimation

8.1 Maximum Likelihood Estimation

The parameters $\boldsymbol{\theta} = (a, b, \sigma)$ can be estimated by maximizing the log-evidence:

$$\boxed{\hat{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta}} \sum_{t=1}^{T} \ln Z_t(\boldsymbol{\theta}).} \tag{52}$$

8.2 Position-Only Log-Likelihood

$$\boxed{\ln L(\boldsymbol{\theta}) = -\frac{1}{2}\sum_{t=1}^{T}\left[\ln\left(2\pi(\Sigma_{11,t}^-(\boldsymbol{\theta}) + \sigma_y^2)\right) + \frac{(Y_t - \mu_{x,t}^-(\boldsymbol{\theta}))^2}{\Sigma_{11,t}^-(\boldsymbol{\theta}) + \sigma_y^2}\right].} \tag{53}$$

8.3 Bayesian Parameter Estimation

The posterior distribution of parameters is:

$$\boxed{p(\boldsymbol{\theta}|\mathbf{Y}_{1:T}) \propto p(\boldsymbol{\theta}) \prod_{t=1}^{T} Z_t(\boldsymbol{\theta}).} \tag{54}$$

9 Information-Theoretic Measures

9.1 Information Gain

The entropy of a Gaussian distribution is:

$$S(\Sigma) = \frac{1}{2} \ln \left[(2\pi e)^2 \det \Sigma \right]. \tag{55}$$

The information gain from a measurement is:

$$\boxed{\Delta I_t = S(\Sigma_t^-) - S(\Sigma_t^+) = \frac{1}{2} \ln \frac{\det \Sigma_t^-}{\det \Sigma_t^+} \geq 0.} \tag{56}$$

9.2 Mutual Information

The mutual information between the state and the measurement is:

$$\boxed{I(\mathbf{X}_t; \mathbf{Y}_t) = \frac{1}{2} \ln \frac{\det(H\Sigma_t^- H^T + R)}{\det R} \geq 0.} \tag{57}$$

9.3 KL Divergence

The KL divergence between prior and posterior is:

$$\boxed{\mathrm{KL}(p^+||p^-) = \frac{1}{2} \left[\ln \frac{\det \Sigma^-}{\det \Sigma^+} - 2 + \mathrm{Tr}((\Sigma^-)^{-1}\Sigma^+) + (\boldsymbol{\mu}^+ - \boldsymbol{\mu}^-)^T (\Sigma^-)^{-1} (\boldsymbol{\mu}^+ - \boldsymbol{\mu}^-) \right].} \tag{58}$$

This equals the information gain ΔI_t when the mean does not change.

10 Special Cases

10.1 Perfect Measurements ($R \to 0$)

As measurement noise vanishes:

$$\Sigma_t^+ \to 0, \quad \boldsymbol{\mu}_t^+ \to \mathbf{Y}_t. \tag{59}$$

The information gain diverges as:

$$\Delta I_t \sim -\frac{1}{2} \ln \det R \to \infty. \tag{60}$$

10.2 Noisy Measurements ($R \to \infty$)

As measurement noise becomes infinite:

$$K_t \to 0, \quad \Sigma_t^+ \to \Sigma_t^-, \quad \boldsymbol{\mu}_t^+ \to \boldsymbol{\mu}_t^-. \tag{61}$$

The information gain vanishes as:

$$\Delta I_t \sim \frac{1}{2}\operatorname{Tr}((\Sigma_t^-)^{-1} H^T R^{-1} H) \to 0. \tag{62}$$

10.3 Stationary Filter

In the stationary regime, the Kalman gain converges to a constant:

$$\boxed{K_{\text{stat}} = \Sigma_{\text{stat}} H^T (H \Sigma_{\text{stat}} H^T + R)^{-1},} \tag{63}$$

where

$$\Sigma_{\text{stat}} = \begin{pmatrix} \frac{\sigma^2}{2ab} & 0 \\ 0 & \frac{\sigma^2}{2a} \end{pmatrix}. \tag{64}$$

For position-only measurements:

$$K_{\text{stat}} = \frac{1}{\frac{\sigma^2}{2ab} + \sigma_y^2} \begin{pmatrix} \frac{\sigma^2}{2ab} \\ 0 \end{pmatrix}. \tag{65}$$

11 Summary of Key Results

Table 1: The Complete Filtering Equations

Step	**Equation**
Predicted Mean	$\boldsymbol{\mu}_t^- = \Phi \boldsymbol{\mu}_{t-1}^+$
Predicted Covariance	$\Sigma_t^- = \Phi \Sigma_{t-1}^+ \Phi^T + Q(\Delta t)$
Innovation	$\mathbf{r}_t = \mathbf{Y}_t - H\boldsymbol{\mu}_t^-$
Innovation Covariance	$S_t = H\Sigma_t^- H^T + R$
Kalman Gain	$K_t = \Sigma_t^- H^T S_t^{-1}$
Updated Mean	$\boldsymbol{\mu}_t^+ = \boldsymbol{\mu}_t^- + K_t \mathbf{r}_t$
Updated Covariance	$\Sigma_t^+ = (I - K_t H)\Sigma_t^-$
Evidence	$Z_t = \mathcal{N}(\mathbf{Y}_t; H\boldsymbol{\mu}_t^-, S_t)$

12 Conclusion

We have developed a complete Bayesian framework for state estimation of the stochastic damped harmonic oscillator. The key results are:

1. **Gaussian Conjugacy:** The prior, likelihood, and posterior are all Gaussian distributions, leading to exact closed-form update equations.

Table 2: Position-Only Measurement Update

Quantity	**Formula**
S_t	$\Sigma^-_{11,t} + \sigma_y^2$
K_t	$\frac{1}{S_t}\begin{pmatrix}\Sigma^-_{11,t}\\ \Sigma^-_{12,t}\end{pmatrix}$
$\mu^+_{x,t}$	$\mu^-_{x,t} + \frac{\Sigma^-_{11,t}}{S_t}(Y_t - \mu^-_{x,t})$
$\mu^+_{v,t}$	$\mu^-_{v,t} + \frac{\Sigma^-_{12,t}}{S_t}(Y_t - \mu^-_{x,t})$
$\Sigma^+_{11,t}$	$\frac{\Sigma^-_{11,t}\sigma_y^2}{S_t}$
$\Sigma^+_{12,t}$	$\frac{\Sigma^-_{12,t}\sigma_y^2}{S_t}$
$\Sigma^+_{22,t}$	$\Sigma^-_{22,t} - \frac{(\Sigma^-_{12,t})^2}{S_t}$

Table 3: Stationary Limits

Quantity	**Value**
Σ_{stat}	$\begin{pmatrix}\frac{\sigma^2}{2ab} & 0\\ 0 & \frac{\sigma^2}{2a}\end{pmatrix}$
K_{stat} (position)	$\frac{1}{\frac{\sigma^2}{2ab}+\sigma_y^2}\begin{pmatrix}\frac{\sigma^2}{2ab}\\ 0\end{pmatrix}$
Σ^+_{stat} (position)	$\begin{pmatrix}\frac{\sigma^2}{2ab}\frac{\sigma_y^2}{\frac{\sigma^2}{2ab}+\sigma_y^2} & 0\\ 0 & \frac{\sigma^2}{2a}\end{pmatrix}$

2. **Explicit Prediction Step:** The prior mean and covariance are obtained by propagating the previous posterior through the system dynamics:

$$\boldsymbol{\mu}_t^- = \Phi\boldsymbol{\mu}_{t-1}^+, \quad \Sigma_t^- = \Phi\Sigma_{t-1}^+\Phi^T + Q(\Delta t).$$

3. **Explicit Update Step:** The posterior mean and covariance are obtained via the Kalman filter equations:

$$K_t = \Sigma_t^- H^T(H\Sigma_t^- H^T + R)^{-1},$$

$$\boldsymbol{\mu}_t^+ = \boldsymbol{\mu}_t^- + K_t(\mathbf{Y}_t - H\boldsymbol{\mu}_t^-),$$
$$\Sigma_t^+ = (I - K_t H)\Sigma_t^-.$$

4. **Explicit Formulas:** We provided complete formulas for position-only, velocity-only, and full-state measurements.

5. **Bayesian Evidence:** We derived the marginal likelihood for model selection and parameter estimation.

6. **Information-Theoretic Measures:** We derived the information gain, mutual information, and KL divergence.

This framework provides a rigorous mathematical foundation for data assimilation in stochastic dynamical systems, with applications in physics, engineering, and finance.

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A Matrix Inversion Lemma

The matrix inversion lemma (Woodbury identity) states that for invertible matrices A and C:

$$(A + BC^{-1}B^T)^{-1} = A^{-1} - A^{-1}B(C + B^T A^{-1}B)^{-1}B^T A^{-1}.$$

This is used to derive:

$$\Sigma_t^+ = \left[(\Sigma_t^-)^{-1} + H^T R^{-1} H\right]^{-1} = \Sigma_t^- - \Sigma_t^- H^T(H\Sigma_t^- H^T + R)^{-1}H\Sigma_t^-.$$

B Derivation of Posterior Mean

From (22) and (23):

$$\boldsymbol{\mu}_t^+ = \Sigma_t^+\left[(\Sigma_t^-)^{-1}\boldsymbol{\mu}_t^- + H^T R^{-1}\mathbf{Y}_t\right].$$

Using the matrix inversion lemma:

$$\Sigma_t^+ = \Sigma_t^- - \Sigma_t^- H^T(H\Sigma_t^- H^T + R)^{-1}H\Sigma_t^-.$$

Let $S = H\Sigma_t^- H^T + R$. Then:

$$\boldsymbol{\mu}_t^+ = \Sigma_t^-(\Sigma_t^-)^{-1}\boldsymbol{\mu}_t^- + \Sigma_t^- H^T R^{-1}\mathbf{Y}_t - \Sigma_t^- H^T S^{-1} H\Sigma_t^-(\Sigma_t^-)^{-1}\boldsymbol{\mu}_t^- - \Sigma_t^- H^T S^{-1} H\Sigma_t^- H^T R^{-1}\mathbf{Y}_t.$$

$$= \boldsymbol{\mu}_t^- + \Sigma_t^- H^T R^{-1}\mathbf{Y}_t - \Sigma_t^- H^T S^{-1} H\boldsymbol{\mu}_t^- - \Sigma_t^- H^T S^{-1}(H\Sigma_t^- H^T)R^{-1}\mathbf{Y}_t.$$

Since $H\Sigma_t^- H^T = S - R$:

$$= \boldsymbol{\mu}_t^- + \Sigma_t^- H^T R^{-1}\mathbf{Y}_t - \Sigma_t^- H^T S^{-1} H\boldsymbol{\mu}_t^- - \Sigma_t^- H^T S^{-1}(S - R)R^{-1}\mathbf{Y}_t.$$

$$= \boldsymbol{\mu}_t^- + \Sigma_t^- H^T R^{-1}\mathbf{Y}_t - \Sigma_t^- H^T S^{-1} H\boldsymbol{\mu}_t^- - \Sigma_t^- H^T S^{-1}\mathbf{Y}_t + \Sigma_t^- H^T R^{-1}\mathbf{Y}_t.$$

Using the simpler derivation from Section 5.3:

$$\boldsymbol{\mu}_t^+ = \boldsymbol{\mu}_t^- + \Sigma_t^+ H^T R^{-1}(\mathbf{Y}_t - H\boldsymbol{\mu}_t^-).$$

Using $\Sigma_t^+ H^T R^{-1} = \Sigma_t^- H^T(H\Sigma_t^- H^T + R)^{-1} = K_t$:

$$\boldsymbol{\mu}_t^+ = \boldsymbol{\mu}_t^- + K_t(\mathbf{Y}_t - H\boldsymbol{\mu}_t^-).$$

C The Stationary Kalman Filter

The stationary Kalman filter arises when $\Sigma_t^- \to \Sigma_{\text{stat}}^-$ and $\Sigma_t^+ \to \Sigma_{\text{stat}}^+$. The stationary covariance satisfies:

$$\Sigma_{\text{stat}}^- = \Phi\Sigma_{\text{stat}}^+\Phi^T + Q(\Delta t),$$
$$\Sigma_{\text{stat}}^+ = \Sigma_{\text{stat}}^- - \Sigma_{\text{stat}}^- H^T(H\Sigma_{\text{stat}}^- H^T + R)^{-1}H\Sigma_{\text{stat}}^-.$$

For $\Delta t \to \infty$, $\Phi \to 0$ and $Q(\infty) \to \Sigma_{\text{stat}}$, so:

$$\Sigma_{\text{stat}}^- = \Sigma_{\text{stat}}.$$

The stationary Kalman gain is:

$$K_{\text{stat}} = \Sigma_{\text{stat}} H^T(H\Sigma_{\text{stat}} H^T + R)^{-1}.$$

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